First Blind Attitude Inversion from Specular Glint Constraints

Weekly Report — 27 March 2026

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Summary

Working end-to-end blind inversion pipeline. From a single observed light curve, recovers ω direction to $\sim 2°$ and magnitude to $< 0.2\%$ in **5–7 minutes**. Tested on 6 trajectories: 4/6 succeed.

Best result (seed 93)	ω dir $= 1.25°$, $\lvert\omega\rvert$ err $= 0.01\%$
Known limitation	180° attitude ambiguity (satellite N/S symmetry)
Failure mode	< 10 constraint epochs $\Rightarrow$ insufficient observability

Pipeline

Steps 1–4 are purely geometric (no BRDF/shadows). Step 5 invokes the full ray-traced forward model.

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STEP 1: Peak detection
  peaks = find_peaks(observed_lc, prominence > 0.3)
  |w|_est = 0.040 * n_peaks + 0.042 deg/s (rho=0.92, 13% err)
  specular = peaks where mag < 6 (guaranteed +/-X alignment with PAB)
  bright = peaks where 6 < mag < 9 (some normal aligned with PAB)
  anchor = brightest specular glint -> 1-DOF circle in phi

STEP 2: Omega grid search [~100s, 16 cores]
  for each of 2000 dirs x 20 mags (|w|_est +/- 20%):
    propagate identity quat from anchor -> delta_q at constraints
    sweep 36 phi values x {+X, -X} anchors (vectorised)
    score: C = w_s * sum(1 - align_pmX)^2 at specular epochs
             + w_b * sum(1 - align_any)^2 at bright epochs
    keep best (phi, cost) per direction

STEP 3: Nelder-Mead refinement [~14s, 16 cores]
  refine top 20 grid results (3D omega, 360 fine phi bins)
  extract top 5 omegas x {+X, -X} best phi = 10 candidates
  back-propagate each to t=0

STEP 4: Geometric refinement + cluster detection [~63s, 8 cores]
  L-BFGS-B on all 10 candidates (specular + bright alignment, no BRDF)
  sort by geo cost -> find first 10x gap -> low-cost cluster

STEP 5: Hi-fi LC scoring [~150s, 8 cores]
  full ray-traced LC for cluster members
  rank by MSE residual -> winner
```

Step	What	Time
1	Peak detection + constraints	< 1 s
2	Grid search (2000×20, 36 ϕ, vectorised)	98 s
3	NM refinement (top 20, 360 ϕ)	14 s
4	Geometric refinement + cluster detection	63 s
5	Hi-fi LC (cluster members)	153 s
	Total	**5.5 min**

Key ideas:

- **Delta-q factorisation:** propagation is q-independent for torque-free dynamics. One propagation per ω serves all 360 ϕ candidates.
- **Specular $\Rightarrow$ $\pm X$:** every mag < 6 peak is a $\pm X$ face alignment (100% across 6000+ peaks). This constrains attitude to a 1-DOF circle per glint.
- **Cluster detection:** geometric cost separates viable candidates from junk (typically 10–75$\times$ gap). Only the cluster gets the expensive hi-fi evaluation.

Candidate Disambiguation

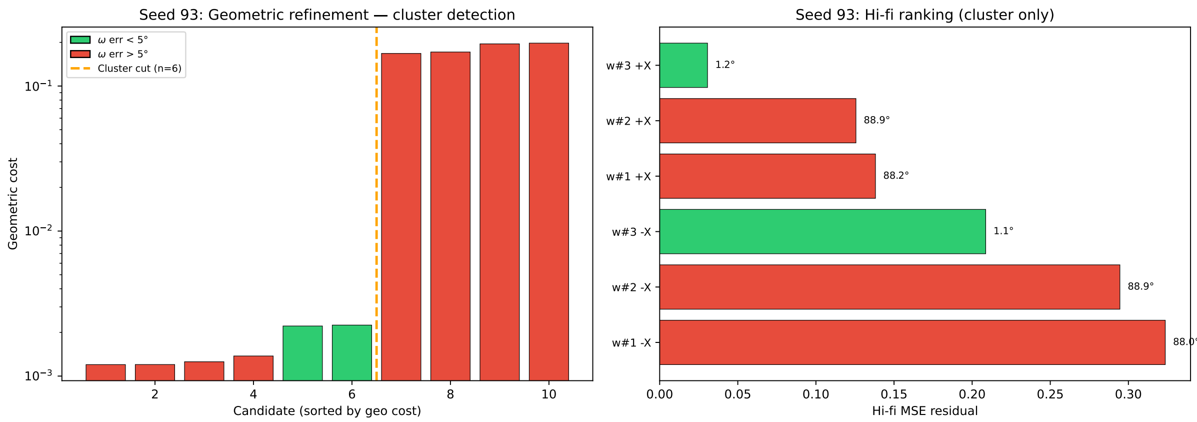

Figure 1: Seed 93. Left: geometric cost (log scale) — 6 candidates below the 75$\times$ gap. Right: hi-fi residual ranks correct ω at #1 with 4$\times$ margin.

Geometric Degeneracy

IS-901 has **mirror symmetry about the body XY plane**: the north and south solar panels are identical, and the bus is symmetric above and below.

- For any ϕ, the attitude at $\phi + 180°$ swaps N/S panels $\Rightarrow$ nearly identical satellite geometry and light curve
- ω recovery is unaffected (both ϕ values yield the same ω)
- q_0 differs by $\sim 180°$ between the two
- This is a **true geometric degeneracy** of the satellite shape — not a pipeline limitation
- **Fix:** sample $\phi \in [0°, 180°)$ instead of $[0°, 360°)$. This eliminates the duplicate and halves the phi search space with no loss of information.

Note: the $\pm X$ anchor ambiguity (front vs back of the bus) is a *separate* question. The front and back of IS-901 are *not* symmetric (different dish/antenna geometry), so $\pm X$ disambiguation is feasible via hi-fi residual comparison. The pipeline already handles both $\pm X$ anchors internally.

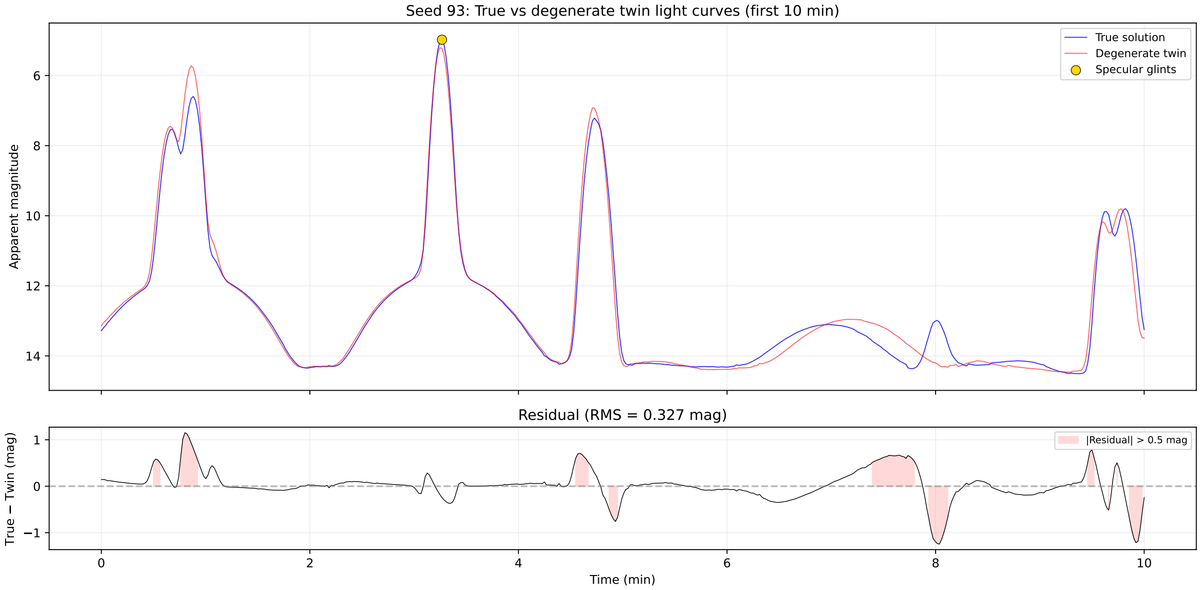

Figure 2: True (blue) vs degenerate twin (red, 179° offset, same ω). Specular glints (gold) match by construction; differences appear in shadow-driven dim epochs.

Multi-Seed Results

6 trajectories, $|\omega| \in [0.85, 1.45]$ deg/s, ≥ 3 specular glints each.

Table 1: $n_{\rm sp}$: specular glints (incl. anchor). $n_{\rm br}$: bright peaks. $n_{\rm con} = (n_{\rm sp}-1) + n_{\rm br}$: constraint epochs.

Seed	$\lvert\omega\rvert$ (deg/s)	$n_{\rm sp}$	$n_{\rm br}$	$n_{\rm con}$	Cluster	ω err (°)	q_0 err (°)	$\lvert\omega\rvert$ err (%)	Time (min)
93	1.28	7	12	18	6	**1.3**	178.9	−0.0	5.5
0	1.45	4	7	10	2	**1.7**	4.4	+0.1	4.5
74	1.10	4	9	12	10	**2.0**	178.6	−0.1	7.4
14	1.23	3	11	11	10	**2.2**	5.2	−0.0	6.7
36	1.43	4	5	8	10	5.5	176.3	+0.2	6.6
27	1.14	4	5	8	10	42.8	129.8	−9.6	6.7

- **4/6 succeed** ($\omega < 2.2°$, $|\omega|$ err $< 0.2\%$). All have $n_{\rm con} \geq 10$.
- **2/6 fail** — both have $n_{\rm con} = 8$. Threshold appears to be $\sim$10 constraint epochs.
- Seed 14: only 3 specular, but 11 bright compensates ($n_{\rm con} = 11$).
- 180° phi ambiguity in 4/6 seeds. Hi-fi resolves it in 2 cases (seeds 0, 14: $q_0 < 6°$).
- Runtime: 4.5–7.4 min.

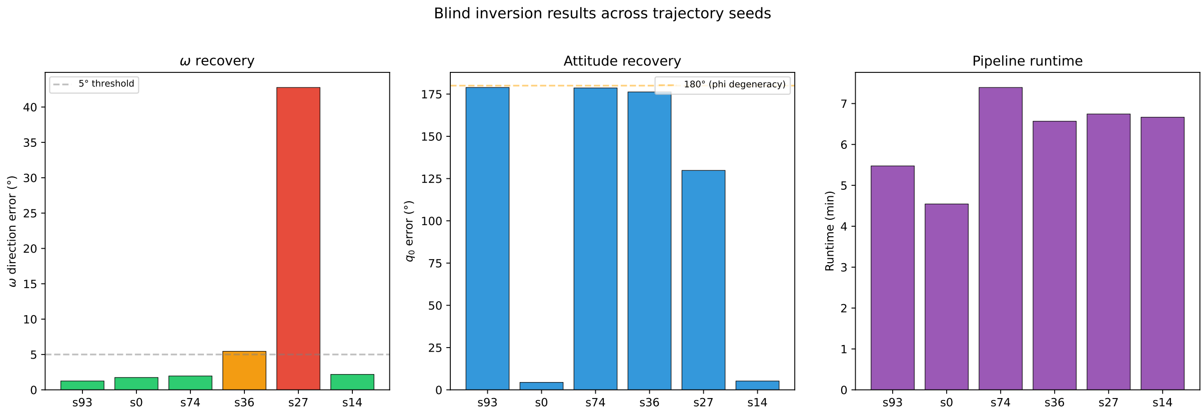

Figure 3: ω direction error, attitude error, and runtime across seeds. Green: $< 5°$. Orange: 5–20°. Red: $> 20°$.

Next Steps

- Restrict $\phi \in [0°, 180°)$ to eliminate the N/S degeneracy (halves candidate count)
- Larger seed sample (≥ 20 trajectories)
- Investigate failure mode for $n_{\text{con}} < 10$: denser grid or additional constraint types?
- Tune hand-picked parameters across the multi-seed set: alignment weights ($w_s = 10$, $w_b = 5$), NM candidate count (top 5 is arbitrary), grid density (2000 dirs, 20 mags — magnitude axis is cheap to densify; ∼0.5% spacing (∼80 bins) costs little extra since propagation time is dominated by direction, not magnitude rescaling)
- Sensitivity to noise level and observation window length